

Physics for EEE 2022 scheme

Question Bank

Module-4 (Maxwell's Equations and EM waves)

1. Explain scalar product and vector product.
2. Explain vector operator Grad with an example.
3. Describe the concepts of divergence, Give its physical significance.
4. Describe the concepts curl. Give its physical significance.
5. Explain the features of Linear, Surface and Volume integrals.
6. Explain briefly the Gauss law in electrostatics and also in magnetism.
7. Explain Ampere's circuital law. Give its differential equation.
8. State and explain Biot-Savart's law.
9. Explain Faraday's law, derive the equation in differential form in case of time varying field.
10. State and Prove Gauss divergence theorem.
11. Mention stokes theorem.
12. Discuss continuity equation.
13. Write a note on modified Ampere's law.
14. Define displacement current. Derive an expression for Displacement current.
15. List four Maxwell's equations for both time varying and static conditions.
16. What is EM wave? Show that EM wave travels with the speed of light.
17. Derive the expression for wave equation for electromagnetic waves for free space using Maxwell's equations.
18. Explain Linear, Circular and Elliptical polarizations of EM waves with diagrams.